

## Review

# Life cycle complexity drives variation in thermal tolerance and plasticity

Patrice Pottier<sup>1,2,3</sup>, Vanessa Kellermann<sup>4,\*,#</sup>,  
Daniel WA Noble<sup>2,\*,†</sup>, Carla M Sgrò<sup>5,\*,‡</sup>,  
John S Terblanche<sup>6,\*,¶</sup> and Belinda van Heerwaarden<sup>4,7,\*,||</sup>



Accumulating evidence suggests that heat tolerance varies substantially across insect development, yet patterns of variation remain difficult to generalise across species. We discuss how the diversity of insect developmental strategies shapes both the intensity and predictability of thermal environments across ontogeny, and how this likely generates variation in heat tolerance, plasticity and carry-over effects. We hypothesise that large developmental variation is expected in holometabolous insects and in species undergoing pronounced microhabitat or diel activity transitions. These transitions can modify heat exposure, behavioural thermoregulatory abilities and the physiological or genetic regulatory network underlying heat tolerance, weakening correlations among life stages. We discuss when carry-over effects are likely to be adaptive, highlighting the importance of environmental predictability, ecological similarity among stages and the balance between heat injury and repair. We argue that an ontogenetic perspective capturing the microenvironmental conditions experienced by each life stage is essential for predicting insect vulnerability to extreme heat.

## Addresses

<sup>1</sup> Department of Biological and Environmental Sciences, Faculty of Science, University of Gothenburg, Gothenburg, Sweden

<sup>2</sup> Division of Ecology and Evolution, Research School of Biology, The Australian National University, Canberra, Australian Capital Territory, Australia

<sup>3</sup> Evolution & Ecology Research Centre, School of Biological, Earth and Environmental Sciences, University of New South Wales, Sydney, New South Wales, Australia

<sup>4</sup> School of Agriculture, Biomedicine and Environment, La Trobe University, Bundoora, Victoria, Australia

<sup>5</sup> School of Biological Sciences, Monash University, Melbourne, VIC, Australia

<sup>6</sup> Centre for Invasion Biology, Department of Conservation Ecology & Entomology, Faculty of AgriSciences, Stellenbosch University, Stellenbosch, South Africa

<sup>7</sup> School of BioSciences, Bio21 Molecular Science and Biotechnology Institute, The University of Melbourne, Parkville, Australia

Corresponding author: Pottier, Patrice ([patrice.pottier@bioenv.gu.se](mailto:patrice.pottier@bioenv.gu.se))

\* Equal contributions

# <https://orcid.org/0000-0002-9859-9642>

† <https://orcid.org/0000-0001-9460-8743>

‡ <https://orcid.org/0000-0001-7950-2246>

¶ <https://orcid.org/0000-0001-9665-9405>

|| <https://orcid.org/0000-0003-2435-2900>

Current Opinion in Insect Science 2026, 76:101523

This review comes from a themed issue on **Global change biology**

Edited by **Sarah Diamond, Ryan Martin, Mike Moore and Carmen da Silva**

Available online 6 April 2026

<https://doi.org/10.1016/j.cois.2026.101523>

2214–5745/© 2026 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

## Introduction

Climate change is a dominant driver of global insect declines [1–4]. Because temperature governs nearly every aspect of insect physiology, ongoing climate warming and increasingly frequent and intense heat extremes are expected to accelerate insect declines [2,4]. Understanding why some species are more sensitive to heat stress than others, and predicting how warming will affect natural insect populations, remains a central challenge in global change biology.

Accumulating evidence indicates that heat tolerance—here defined as the likelihood that an organism approaches or exceeds its upper thermal limits—is not uniform across an insect's life cycle [5–8]. Developmental stages can differ markedly in their upper thermal limits and plasticity, reflecting stage-specific differences in morphology, physiology, mobility and microhabitat use [9–11]. However, trends in heat tolerance variation among life stages remain difficult to generalise across insects. While some studies report higher sensitivity to heat stress in embryos or first larval instars (e.g. [12–15]), others identify heightened sensitivity during the pupal or adult stage (e.g. [15,16]). Developmental variation in heat tolerance within a species can also be comparable or greater in magnitude to differences among species [7].

**Box 1 Plasticity within and across life stages.**

Insects differ widely in how they experience thermal environments across their life cycle, creating variation in the opportunities for selection on heat tolerance and plasticity [17]. Plasticity — the ability of an organism to modify its phenotype in response to environmental conditions — is itself an evolvable trait and may occur within life stages (e.g. hardening, acclimation), across successive life stages (developmental acclimation), or across generations [18–20]. Carry-over effects describe the persistence of environmental effects across life stages and can lead to transgenerational effects; however, we focus here on carry-over effects across life stages.

Theoretical models predict that frequent exposure to temperatures approaching physiological limits should favour selection for high, canalised heat tolerance, whereas plasticity should be favoured in environments that are variable but predictable (i.e. high temporal autocorrelation [21,22]) and when the costs of plasticity are low [23,24]. With respect to insect life cycles, stages that are most consistently exposed to extreme heat should evolve higher heat tolerance, while stages experiencing variable, predictable thermal variation should exhibit greater within-stage plasticity. Plasticity can be reversible or irreversible, depending on the underlying physiological mechanisms and the predictability of environmental conditions. Reversible plasticity is expected to be favoured in stages experiencing short-term or fluctuating conditions, and irreversible, developmentally fixed plasticity favoured when conditions are stable and predictable across life stages [18,23,24]. When thermal environments differ substantially between stages (e.g. due to shifts in microhabitat, behaviour, or timing of exposures), developmental plasticity may be of limited value and even hamper evolutionary responses [25].

Life-cycle complexity can shape the dynamics of heat stress responses by influencing both the thermal variability experienced within stages and the continuity of conditions experienced across stages. Longer-lasting stages, or stages that are exposed to seasonal variation, often experience a broader range of temperatures, increasing opportunities for reversible acclimation. In contrast, short-lasting stages or those confined to narrow seasonal windows may experience limited thermal variation and predictability, promoting canalisation of thermal traits. Phenology and voltinism also intricately contribute to the form and extent of thermal exposure. For instance, eggs, larvae and adults may occur across distinct seasons, particularly for uni- and bivoltine species, meaning not all life stages may be exposed to heat extremes. In contrast, in multivoltine species, all life stages may occur within the same season, effectively spreading heat stress and selection for heat tolerance across development, although different stages may nonetheless experience different microclimates.

Behavioural thermoregulatory abilities also play an important role in modulating the magnitude and predictability of experienced thermal conditions across life stages. Mobile stages, such as flying adults, may have more opportunities to escape heat extremes, weakening selection for high heat tolerance and plasticity (i.e. Bogert effect [26,27]). In contrast, immobile stages, such as embryos and pupae, are forced to cope with local microenvironmental conditions, which can generate strong selection for high heat tolerance and plasticity in microsites exposed to high thermal variability.

This framework predicts that different insect life cycles, by shaping the timing, duration, variability and predictability of heat exposure, will determine whether selection favours canalised heat tolerance, reversible within-stage plasticity, or adaptive carry-over effects across life stages (Figure 1).

Here, we provide a conceptual framework for how complex insect life cycles can result in life stages experiencing different levels of exposure to heat stress (i.e. different mean, variance and predictability) across development, and how this can lead to evolved differences in heat tolerance and plasticity within and across life stages (Box 1). While some of these points have been discussed previously [5–8], we hope to highlight recent case studies that add to these discussions.

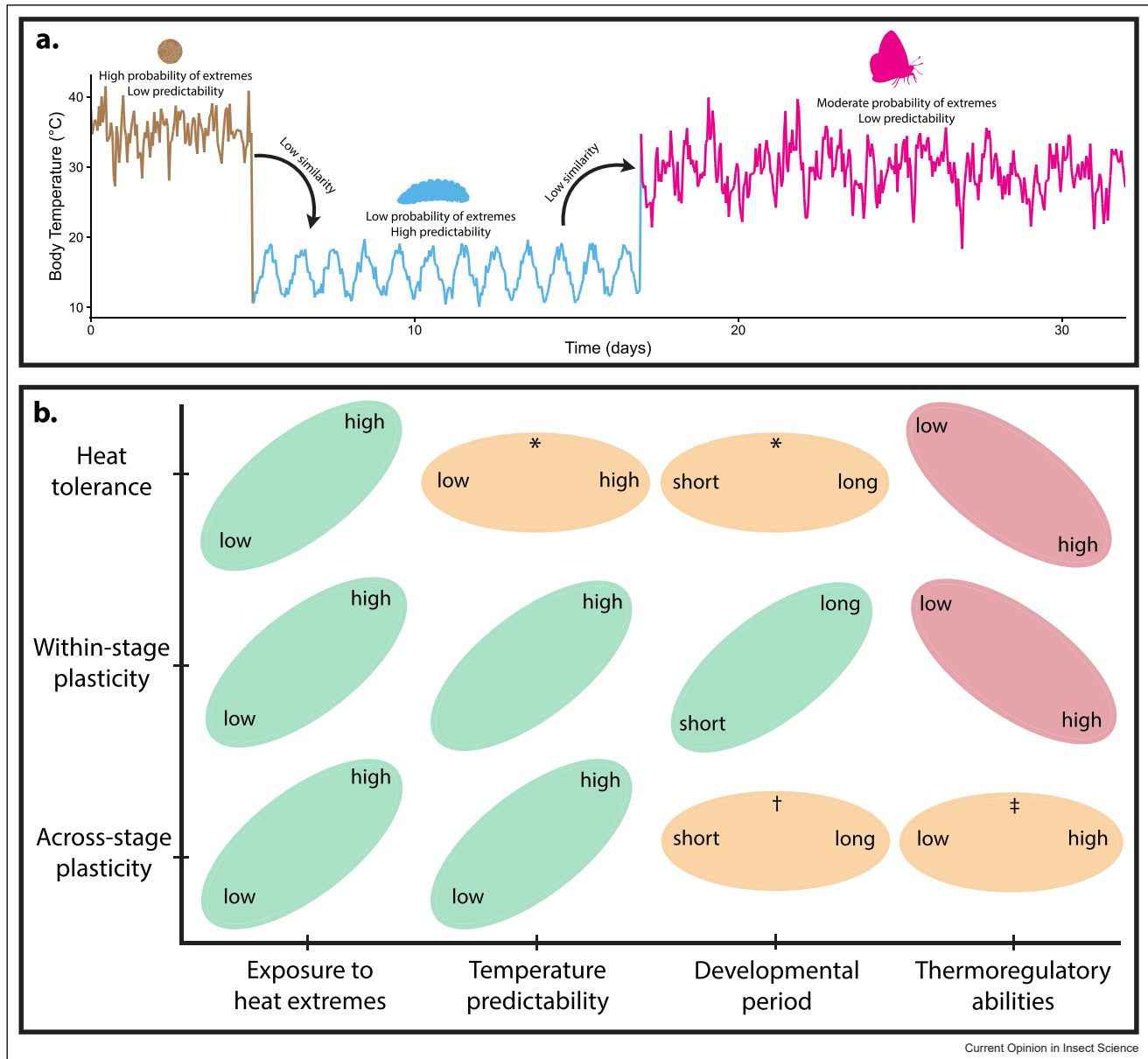
### Why complex life cycles can promote variation in heat tolerance and plasticity

#### Holometaboly and hemimetaboly

Holometabolous insects undergo complete metamorphosis, whereas hemimetabolous juveniles resemble small adults and lack a pupal stage (Figure 2). Given that microclimatic conditions can differ substantially between life stages and be less predictable in holometabolous insects, we predict developmental differences in heat tolerance and plasticity to be generally more common than in hemimetabolous insects. There is some evidence to support this hypothesis. In the moth *Manduca sexta*, for example, larvae experience more variable temperatures and lower humidity than eggs or pupae, which may explain their higher heat tolerance [6]. In

another study, pupae of the butterfly *Bicyclus anynana* exhibit high heat tolerance compared to eggs, consistent with their development within the humid, thermally buffered boundary layer of plant leaves [12]. In contrast, many hemimetabolous insects (e.g. Orthoptera, Hemiptera, Blattodea) typically maintain similar microhabitats and activity patterns across juvenile and adult stages, with the embryonic stage being the main exception. This continuity in microclimatic niche is accompanied by more gradual changes in body size across development compared with holometabolous insects. Because surface area-to-volume ratio strongly influences rates of water loss and heat exchange, these smaller size transitions are expected to generate more modest shifts in heat tolerance across life stages. Nevertheless, body size can also shape heat tolerance within holometabolous insects even when microclimatic niches remain similar, such as across larval instars, due to variation in body size [12]. In addition, hemimetabolous insects from the class Odonata (dragonflies, damselflies), Ephemeroptera (mayflies) and Plecoptera (stoneflies), for example, undergo a significant aquatic-to-terrestrial microhabitat transition, and are thus likely to experience pronounced variation in microclimate exposure across their life cycle (Figure 3).

Figure 1

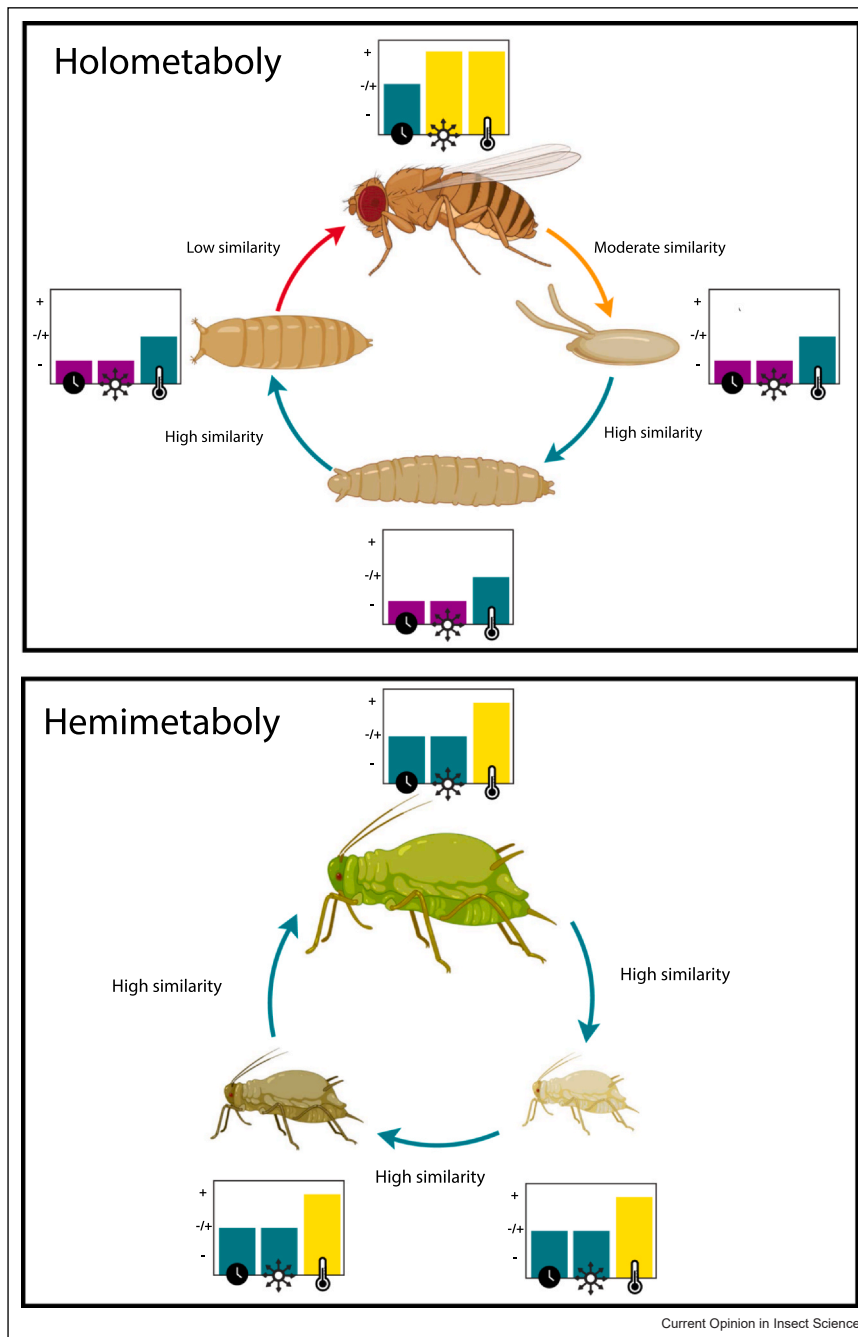


Conceptual predictions for variation in heat tolerance and plasticity within and across life stages. **(a)** Example of variation in thermal exposure across the life cycle of a holometabolous insect. Displayed are simulated data varying in mean temperature, variance and predictability among life stages, resulting in differences in ecological similarity between life stages. **(b)** Conceptual predictions. Green ellipses indicate positive associations between factors, red ellipses indicate negative associations and orange ellipses denote context-dependent responses. For example, high temperature predictability or short developmental windows are not necessarily associated with high heat tolerance if operative body temperatures do not approach tolerance limits (\*). Similarly, whether developmental plasticity carries over across life stages depends on variation in developmental periods (†) and thermoregulatory abilities (‡) among stages and which stages are compared.

A recent meta-analysis has found greater plasticity in early life stages compared to adults, and greater plasticity in hemimetabolous insects relative to holometabolous insects [28]. These patterns are consistent with the idea that many hemimetabolous taxa experience environmental cues that may often be more predictable than

those experienced by holometabolous taxa across their life cycle, potentially favouring the adaptive value of developmental carry-over effects (Figure 2). Variation in plasticity among life stages in holometabolous insects also likely reflects the strong asymmetry in the duration of their developmental stages. Eggs and pupae are likely to

Figure 2

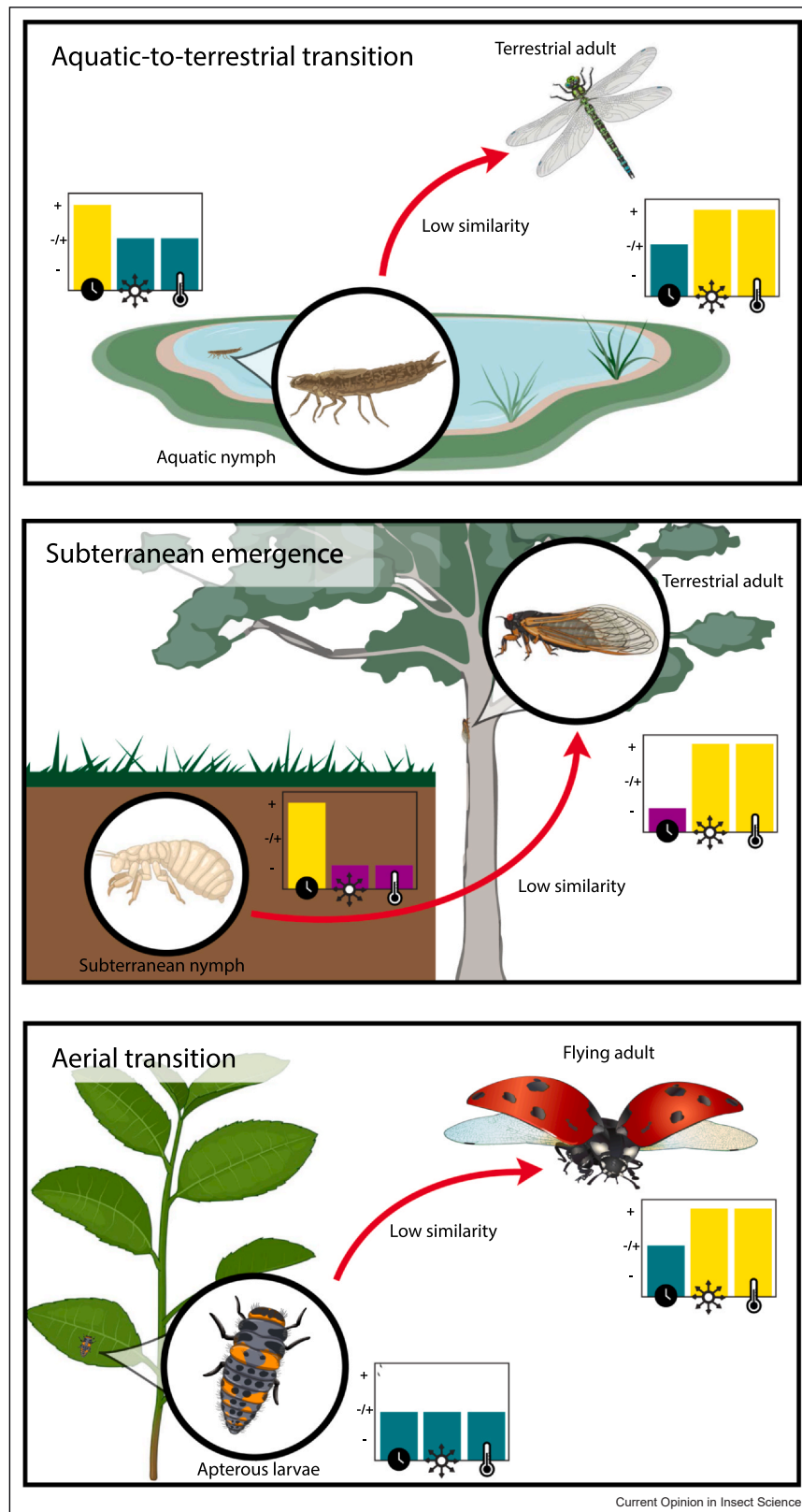


Example of variation in heat stress exposure among life stages in holometabolous and hemimetabolous insects. Vinegar flies (*Drosophila melanogaster*) and green peach aphids (*Myzus persicae*) are used as examples of holometabolous and hemimetabolous insects, respectively. These species illustrate key contrasts between holometabolous and hemimetabolous life cycles, but they are not representative of all insects in these groups, as developmental periods, thermoregulatory abilities and exposure to extreme heat vary strongly among species (e.g. dragonflies and cicadas in Figure 3). Note that only the summer cycle of the green peach aphid is presented here. Variation in developmental periods, thermoregulatory abilities and exposure to heat extremes due to microhabitat differences (left to right) is presented relative to other life stages with three values: low (-), moderate (+/-) and high (+). Arrows connecting different life stages represent the degree of ecological similarity among life stages, which is expected to generate developmental variation in heat tolerance and plasticity. Species icons were taken from BioRender.

experience a narrower range of operative temperatures than the longer larval stage, and these differences in experienced body temperatures may contribute to

explaining differences in plasticity across stages. Nonetheless, developmental carry-over effects have been documented in holometabolous insects [29–31],

Figure 3



(caption on next page)

Example of variation in heat stress exposure among life stages of insects with major microhabitat transitions. Mosaic darter dragonflies (*Aeshna* sp.), periodical cicadas (*Magicicada* sp.) and harlequin ladybirds (*Harmonia* sp.) are used as examples of insects experiencing aquatic-to-terrestrial transitions, subterranean emergence and aerial transitions across their life cycle, respectively. Variation in developmental periods, thermoregulatory abilities, and exposure to heat extremes due to microhabitat differences (left to right) is presented relative to other life stages with three values: low (-), moderate (+/-) and high (+). Red arrows connecting different life stages represent a low degree of ecological similarity among life stages, which is expected to generate developmental variation in heat tolerance and plasticity. Species icons were taken from BioRender.

suggesting that metamorphosis does not universally preclude their occurrence. Evidence for greater plasticity in thermal limits in early life stages [28] is also consistent with stronger selection for within-stage plasticity in stages with limited opportunities for behavioural thermoregulation. However, empirical evidence for the Bogert effect remains mixed (e.g. [27,32,33]), potentially because microclimatic variation at biologically relevant scales for each life stage is difficult to quantify [34]. For instance, cryptic thermal variation at the scale of a single leaf can strongly influence the thermal limits of herbivorous insects [35], and mothers may oviposit in microhabitats that are buffered from large environmental fluctuations [36]. Ultimately, predicting variation in heat tolerance and plasticity requires an understanding of the operative body temperatures experienced by each life stage.

Crucially, physiological or genetic regulatory networks underlying heat tolerance and plasticity thereof can also be decoupled to some extent among life stages, especially in holometabolous insects [5,37,38]. Experimental evolution studies in *Drosophila* have demonstrated that selection on heat tolerance in one life stage does not often result in correlated increases in tolerance in other stages, suggesting partial independence in the underlying physiological or genetic regulatory network [39–41]. This decoupling likely reflects the extensive metamorphic reorganisation characteristic of holometabolous development, as well as systematic differences in thermal exposure and selective pressures across life stages. These findings call for an ontogenetic perspective in comparative and evolutionary studies of insect sensitivity to heat stress, particularly in holometabolous taxa where life-stage transitions are most pronounced.

### Microhabitat transitions

Many insects experience pronounced microhabitat transitions across their life cycle, generating substantial differences in thermal exposure across life stages (Figure 3). Aquatic–terrestrial transitions are a clear example of developmental habitat transitions. In dragonflies and damselflies, for instance, early-life stages develop in aquatic habitats, while adults emerge into far more variable terrestrial conditions. Similarly, many insects develop as larvae in thermally stable microhabitats, such as soil (e.g. cicadas, ground beetles) or plant tissues (e.g. leaf miners, borers), before emerging as adults into microhabitats with larger environmental heterogeneity [42]. In addition, transitions to flying adult stages can also have a pronounced influence

on thermal exposure. Apterous or weakly mobile larvae are often constrained to local thermal conditions, whereas flying adults can behaviourally thermoregulate by selecting favourable microclimates.

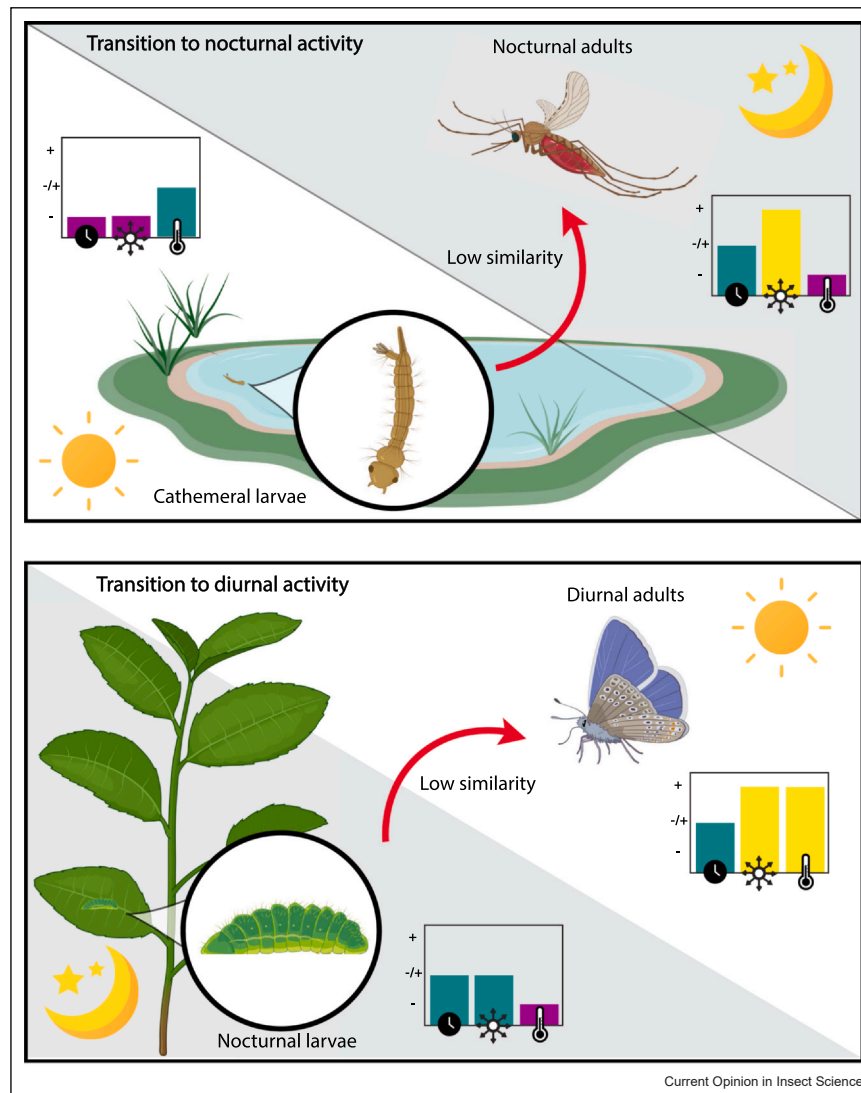
Taken together, differences in microhabitats across life stages can substantially dampen or exacerbate exposure to thermal extremes and thermoregulatory abilities, which is likely to influence selection on heat tolerance and plasticity (Figure 1), such that life stages in more predictable and buffered environments are predicted to have lower and more canalised heat tolerance (Figure 3). There is evidence for lower heat tolerance in thermally buffered (wood-boring or subterranean) larvae relative to flying adults, resulting in lower thermal safety margins [43,44]. However, broad-scale analyses report heterogeneous results regarding differences in thermal plasticity between aquatic and terrestrial insects, suggesting greater plasticity in heat tolerance in aquatic invertebrates [32,45], or little difference between habitats across insects [28]. Nevertheless, within-species comparisons across life stages transitioning to different microhabitats remain largely untested, and it is an interesting avenue for future comparative research. While it will be challenging for some species, we need more studies that characterise the microclimates experienced by different life stages to determine if variation in plastic responses and heat tolerance match theoretical expectations (Figure 1).

### Shifts in diel activity

Transitions between diurnal, crepuscular and nocturnal activity are common across insect life cycles and can substantially alter thermal exposure across development (Figure 4). For example, larvae of many insect species are predominantly nocturnal (e.g. butterflies) or remain concealed during the day within soil (e.g. cicadas), plant (e.g. seed beetles), or host tissues (e.g. parasitoids), whereas adults are often active during the day. In other cases, larvae can develop under cathemeral (i.e. day and night activity) thermal regimes, while adults shift their activity to crepuscular or nocturnal periods (e.g. moths, mosquitoes), reducing exposure to daytime heat extremes. Such shifts can modify not only the mean temperatures experienced, but also the amplitude and predictability of thermal variation across life stages (Figure 4). Therefore, transitions in diel activity are expected to influence selection on heat tolerance and plasticity.

Diurnal stages are more likely to experience frequent exposure to high and variable temperatures, potentially

Figure 4



Example of variation in heat stress exposure among life stages of insects with diel activity transitions. *Anopheles* mosquitoes (*Anopheles* sp.) and common blue butterflies (*Polyommatus* sp.) are used as examples of insects experiencing shifts to nocturnal and diurnal activity across their life cycle, respectively. Variation in developmental periods, thermoregulatory abilities and exposure to heat extremes due to microhabitat differences (left to right) is presented relative to other life stages with three values: low (-), moderate (+/-) and high (+). Red arrows connecting different life stages represent a low degree of ecological similarity among life stages, which is expected to generate developmental variation in heat tolerance and plasticity. Species icons were taken from BioRender.

favouring increased and canalised heat tolerance, whereas nocturnal stages typically encounter cooler and more stable thermal conditions, suggesting lower heat tolerance, but higher within-life stage plasticity. As with microhabitat transitions (see section above), diel shifts are associated with unreliable environmental cues across development and may generate stage-specific heat tolerances that are only weakly correlated. This is evidenced in the fall armyworm, for instance, where 45 °C kills the majority (>80%) of nocturnal adults but does not influence the survival of cathemeral larvae [46]. In addition, adults have an ~4°C lower CT<sub>max</sub> than first-

instar larvae [47]. Similar conclusions have been reached when comparing nocturnal and diurnal bee species [48]. Despite their potential importance, diel activity patterns are not always incorporated into comparative analyses of heat tolerance (but see [49–51]), and may represent an underexplored axis through which life-history complexity shapes insect responses to climate change.

#### Multiple developmental transitions

The developmental strategies highlighted above are not mutually exclusive, and some insects exhibit multiple key transitions across their life cycle. Mosquitoes, for

example, experience complete metamorphosis with pronounced shifts in both microhabitat (aquatic larvae vs. terrestrial flying adults) and diel activity (catemeral larvae vs. crepuscular or nocturnal adults). Such combinations are expected to amplify ontogenetic differences in heat tolerance and potentially weaken the adaptive benefits of carry-over effects across life stages. For example, Orlinick and colleagues [52], recently showed that *Aedes albopictus* diurnal and aquatic larvae have a much higher  $CT_{max}$  than nocturnal, flying adults ( $\Delta CT_{max}$  ranging from 5.6 to 8.4°C). In contrast, a recent study in three hemimetabolous aphid species found a much smaller range of  $CT_{max}$  across life stages ( $\Delta CT_{max}$  = 0.49–0.79°C) [53]. Heat tolerance in adult *Culex pipiens* is also influenced primarily by adult thermal acclimation, with comparatively weak effects of larval conditions [54]. This pattern is consistent with the idea that temperatures experienced during the aquatic larval stage provide an unreliable cue of the thermal conditions encountered by flying adults, limiting the adaptive value of carry-over effects across life stages. Therefore, the degree of ecological similarity among life stages is predicted to be a key determinant of heat tolerance (de)coupling across development.

### Future directions

Although an increasing number of studies demonstrate that insects vary in their heat tolerance and plasticity across life stages, most comparative datasets remain heavily biased towards a single or a few life stages. For example, fewer than 2% of species included in large thermal tolerance compilations, such as GlobTherm [55], have data for both juvenile and adult stages [7]. Embryonic stages are particularly underrepresented, accounting for less than 10% of empirical studies in conservation physiology [56], which likely reflects the logistical challenges of working with small, fragile and rapidly developing eggs. Predicting population-level responses to climate warming will require addressing these gaps through within-species comparisons of heat tolerance and plasticity across life stages.

Predicting the impacts of changing temperatures will also require knowledge of the interplay between thermal damage and repair across all life stages [57,58]. When temperatures fall within the ‘stressful range’ (*sensu* [59]), where damage accumulates faster than repair mechanisms can compensate, developmental exposure may impair heat tolerance, reduce plasticity, or generate latent fitness costs across life stages. This was elegantly shown in flour beetles, where effects on male fertility were most pronounced when heat waves were experienced during the pupal or juvenile stages [16]. Similarly, cold temperatures experienced during the larval stage are most detrimental to female fecundity in seed beetles [60]. In contrast, adult heat stress has more pronounced

impacts on fertility than developmental warming in *Drosophila melanogaster*, because different life stages may also vary in their ability to tolerate different durations of heat stress or repair physiological damage [61]. These apparent carry-over effects can also reflect stage-specific heat sensitivity of reproductive processes. For example, in many insects, spermatogenesis begins during larval or pupal stages, which may be more heat-sensitive than mature sperm, generating delayed impacts on fertility even when stress exposure precedes adulthood [62]. Understanding the physiological mechanisms driving variation in heat tolerance and plasticity across life stages is an important avenue for future research.

Characterising the microenvironments experienced by different life stages, and how these will change in the future, should also be a research priority. Microclimates can vary dramatically at fine spatial scales — even within a single leaf [35] — and these differences can translate into substantial variation in thermal exposure across development [34,63]. In addition, microclimate exposure is expected to vary with insect phenology and life-history traits [64]. Periods of dormancy, such as diapause or quiescence, can coincide with periods of extreme heat, yet how heat tolerance varies between active and dormant stages is largely unknown because most studies have focused on active life stages. In addition, winter warming during diapause can induce strong carry-over effects depending on the life stage overwintering, though this remains rarely studied [65,66]. Therefore, measuring the responses of dormant stages to increased temperatures will be key to assessing developmental variation in heat tolerance and carry-over effects. How voltinism influences variation in heat tolerance across life stages is also not well explored. Although studies in univoltine butterflies suggest that voltinism, through its effects on the timing of thermal exposure during development, is critical in shaping body size, timing of emergence and range shifts [67,68]. The dynamics of thermal acclimation and hardening are also important to consider at fine temporal resolutions. Changes in thermal tolerance can occur within hours in the wild and should be considered [69]. Fine temporal and spatial measurements of the microclimates experienced by each life stage, and how to quantify their influence on phenotypes (e.g. intensity, predictability [22]) will therefore be critical for linking observed heat tolerance, plasticity and produce meaningful estimates of climate vulnerability [34].

### CRedit authorship contribution statement

**Patrice Pottier:** Conceptualization, Visualization, Writing – original draft, Project administration, Writing – review & editing. **Vanessa Kellermann:** Conceptualization, Writing – review & editing. **Daniel W.A. Noble:** Conceptualization, Writing – review & editing. **Carla M. Sgrò:** Conceptualization, Writing –

review & editing. **John S. Terblanche:** Conceptualization, Writing – review & editing. **Belinda van Heerwaarden:** Conceptualization, Writing – review & editing.

## Data Availability

No data were used for the research described in the article.

## Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Patrice Pottier reports financial support was provided by Wenner-Gren Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgements

PP was supported by a Wenner-Gren Stiftelserna postdoctoral fellowship (UPD2024-0239). During the preparation of this manuscript, PP used ChatGPT 5.2 to improve the clarity and grammar of some sentences. The authors reviewed and edited the content as needed and take full responsibility for the content of the published article. We thank Carmen Da Silva, Sarah Diamond, Ryan Martin, and Mike Moore for the invitation to contribute this manuscript.

## References and recommended reading

Papers of particular interest, published within the period of review, have been highlighted as:

- of special interest
- of outstanding interest

1. Farooq M, Khan U, Adnan M, Younas W, Qureshi NB, Yasir QM, Cai Q, Chiu M-C: **Global decline of aquatic and terrestrial insects driven by climate change and anthropogenic impacts: the interaction of multiple stressors and disruption of niche conservatism.** *Biol Conserv* 2025, **308**:111181.
  2. Harvey JA, Tougeron K, Gols R, Heinen R, Abarca M, Abram PK, Basset Y, Berg M, Boggs C, Brodeur J, et al.: **Scientists' warning on climate change and insects.** *Ecol Monogr* 2023, **93**:e1553.
  3. Sockman KW: **Long-term decline in montane insects under warming summers.** *Ecology* 2025, **106**:e70187.
  4. Harvey JA, Heinen R, Gols R, Thakur MP: **Climate change-mediated temperature extremes and insects: from outbreaks to breakdowns.** *Glob Change Biol* 2020, **26**:6685–6701.
  5. Hoffmann AA, Sgrò CM, van Heerwaarden B: **Testing evolutionary adaptation potential under climate change in invertebrates (mostly *Drosophila*): findings, limitations and directions.** *J Exp Biol* 2023, **226**:jeb245749.
  6. Kingsolver JG, Arthur Woods H, Buckley LB, Potter KA, MacLean HJ, Higgins JK: **Complex life cycles and the responses of insects to climate change.** *Integr Comp Biol* 2011, **51**:719–732.
  7. Kingsolver JG, Buckley LB: **Ontogenetic variation in thermal sensitivity shapes insect ecological responses to climate change.** *Curr Opin Insect Sci* 2020, **41**:17–24.
  8. Bowler K, Terblanche JS: **Insect thermal tolerance: what is the role of ontogeny, ageing and senescence?** *Biol Rev* 2008, **83**:339–355.
  9. Marais E, Terblanche JS, Chown SL: **Life stage-related differences in hardening and acclimation of thermal tolerance traits in the kelp fly, *Paractora dreuxi* (Diptera, Helcomyzidae).** *J Insect Physiol* 2009, **55**:336–343.
  10. Belén Arias M, Josefina Poupin M, Lardies MA: **Plasticity of life-cycle, physiological thermal traits and *Hsp70* gene expression in an insect along the ontogeny: effect of temperature variability.** *J Therm Biol* 2011, **36**:355–362.
  11. Mutamiswa R, Machekano H, Chidawanyika F, Nyamukondiwa C: **Life-stage related responses to combined effects of acclimation temperature and humidity on the thermal tolerance of *Chilo partellus* (Swinhoe) (Lepidoptera: Crambidae).** *J Therm Biol* 2019, **79**:85–94.
  12. Klockmann M, Günter F, Fischer K: **Heat resistance throughout ontogeny: body size constrains thermal tolerance.** *Glob Change Biol* 2017, **23**:686–696.
  13. Moghadam NN, Ketola T, Pertoldi C, Bährndorff S, Kristensen TN: **Heat hardening capacity in *Drosophila melanogaster* is life stage-specific and juveniles show the highest plasticity.** *Biol Lett* 2019, **15**:20180628.
  14. Pincebourde S, Casas J: **Warming tolerance across insect ontogeny: influence of joint shifts in microclimates and thermal limits.** *Ecology* 2015, **96**:986–997.
  15. Liu H, Wang X, Chen Z, Lu Y: **Characterization of cold and heat tolerance of *Bactrocera tau* (Walker).** *Insects* 2022, **13**:329.
  16. Sales K, Vasudeva R, Gage MJG: **Fertility and mortality impacts of thermal stress from experimental heatwaves on different life stages and their recovery in a model insect.** *R Soc Open Sci* 2021, **8**:201717.
  17. Woods HA, Pincebourde S, Dillon ME, Terblanche JS: **Extended phenotypes: buffers or amplifiers of climate change?** *Trends Ecol Evol* 2021, **36**:889–898.
  18. West-Eberhard MJ: **Developmental Plasticity and Evolution.** Oxford University Press; 2003.
  19. Chevin L-M, Lande R, Mace GM: **Adaptation, plasticity, and extinction in a changing environment: towards a predictive theory.** *PLoS Biol* 2010, **8**:e1000357.
  20. Sgrò CM, Terblanche JS, Hoffmann AA: **What can plasticity contribute to insect responses to climate change?** *Annu Rev Entomol* 2016, **61**:433–451.
  21. Vasseur DA, Yodzis P: **The color of environmental noise.** *Ecology* 2004, **85**:1146–1152.
  22. Liu M, Bell-Roberts L, Botero CA, Cornwallis CK, West SA: **Environmental predictability in phylogenetic comparative analysis: how to measure it and does it matter?** *Glob Ecol Biogeogr* 2025, **34**:e70108.
- This paper establishes common metrics of environmental predictability for environmental time series, discusses their limitations and evaluates their effectiveness in a comparative context. They show that different measures of environmental predictability can give different results, and so careful thought needs to be given to the choice of metric used to quantify predictability within a study.
23. Leung C, Rescan M, Grulois D, Chevin L-M: **Reduced phenotypic plasticity evolves in less predictable environments.** *Ecol Lett* 2020, **23**:1664–1672.
  24. Gabriel W: **How stress selects for reversible phenotypic plasticity.** *J Evol Biol* 2005, **18**:873–883.
  25. Hoffmann AA, Bridle J: **The dangers of irreversibility in an age of increased uncertainty: revisiting plasticity in invertebrates.** *Oikos* 2022, **2022**:e08715.
  26. Muñoz MM: **The Bogert effect, a factor in evolution.** *Evol* 2022, **76**:49–66.
  27. Marais E, Chown SL: **Beneficial acclimation and the Bogert effect.** *Ecol Lett* 2008, **11**:1027–1036.
  28. Weaving H, Terblanche JS, Pottier P, English S: **Meta-analysis reveals weak but pervasive plasticity in insect thermal limits.** *Nat Commun* 2022, **13**:5292.

29. Kellermann V, Sgrò CM: **Evidence for lower plasticity in CTMAX at warmer developmental temperatures.** *J Evol Biol* 2018, **31**:1300-1312.
  30. Kellermann V, van Heerwaarden B, Sgrò CM: **How important is thermal history? Evidence for lasting effects of developmental temperature on upper thermal limits in *Drosophila melanogaster*.** *Proc Biol Sci* 2017, **284**:20170447.
  31. van Heerwaarden B, Sgrò C, Kellermann VM: **Threshold shifts and developmental temperature impact trade-offs between tolerance and plasticity.** *Proc Biol Sci* 2024, **291**:20232700.
  32. Pottier P, Burke S, Zhang RY, Noble DWA, Schwanz LE, Drobniak SM, Nakagawa S: **Developmental plasticity in thermal tolerance: ontogenetic variation, persistence, and future directions.** *Ecol Lett* 2022, **25**:2245-2268.
  33. Mitchell KA, Sinclair BJ, Terblanche JS: **Ontogenetic variation in cold tolerance plasticity in *Drosophila*: is the Bogert effect bogus?** *Naturwissenschaften* 2013, **100**:281-284.
  34. Kerr JT, Gordon SCC, Chen I-C, Ednie G, Foden W, Newbold T, Reynolds AR, Suggitt AJ, Terblanche JS, Watson MJ: **Effects of microclimate variation on insect persistence under global change.** *Nat Rev Biodivers* 2025, **1**:532-542.
- This perspective paper emphasises the importance of microclimates in alleviating the negative effects of extreme weather events on insects. Microclimate diversity is argued to improve opportunities for adaptive and plastic responses to extremes, thereby allowing populations to persist and the authors argue management plans should promote microclimate heterogeneity to improve conservation outcomes under climate change.
35. Pincebourde S, Casas J: **Narrow safety margin in the phyllosphere during thermal extremes.** *Proc Nat Acad Sci* 2019, **116**:5588-5596.
  36. Raynal R, Pottier P, Nakagawa S, Schwanz L: **Natural developmental temperatures of ectotherms: a systematic map and comparative analysis.** *EcoEvoRxiv* 2025, <https://doi.org/10.32942/X2WS91>
  37. Freda PJ, Alex JT, Morgan TJ, Ragland GJ: **Genetic decoupling of thermal hardiness across Metamorphosis in *Drosophila melanogaster*.** *Integr Comp Biol* 2017, **57**:999-1009.
  38. Freda PJ, Ali ZM, Heter N, Ragland GJ, Morgan TJ: **Stage-specific genotype-by-environment interactions for cold and heat hardiness in *Drosophila melanogaster*.** *Heredity* 2019, **123**:479-491.
  39. Loeschcke V, Krebs RA: **Selection for heat-shock resistance in larval and in adult *Drosophila buzzatii*: comparing direct and indirect responses.** *Evol* 1996, **50**:2354-2359.
  40. Sambucetti P, Scannapieco AC, Loeschcke V, Norry FM: **Heat-stress survival in the pre-adult stage of the life cycle in an intercontinental set of recombinant inbred lines of *Drosophila melanogaster*.** *J Exp Biol* 2013, **216**:2953-2959.
  41. Borda MA, Gomez FH, Sambucetti P, Norry FM: **Genetic variation in the heat-stress survival of embryos is largely decoupled from adult thermotolerance in an intercontinental set of recombinant lines of *Drosophila melanogaster*.** *J Therm Biol* 2021, **102**:103119.
  42. Woods HA, Dillon ME, Pincebourde S: **The roles of microclimatic diversity and of behavior in mediating the responses of ectotherms to climate change.** *J Therm Biol* 2015, **54**:86-97.
  43. Colado R, Sendra A, Pallarés S, Plaza-Buendía J, Velasco J, Sánchez-Fernández D: **Ontogeny shapes vulnerability to climate change underground: larvae of a subterranean beetle are more sensitive to temperature increase than adults.** *Subterr Biol* 2025, **54**:23-33.
  44. Li C, Wang L, Li J, Gao C, Luo Y, Ren L: **Thermal survival limits of larvae and adults of *Sirex noctilio* (Hymenoptera: Siricidae) in China.** *PLoS One* 2019, **14**:e0218888.
  45. Noble DWA, Kar F, Bush A, Seebacher F, Nakagawa S: **Limited plasticity but increased variance in physiological rates across ectotherm populations under climate change.** *Funct Ecol* 2025, **39**:1176-1193.
  46. Tao Y-D, Liu Y, Wan X-S, Xu J, Fu D-Y, Zhang J-Z: **High and low temperatures differentially affect survival, reproduction, and gene transcription in male and female moths of *Spodoptera frugiperda*.** *Insects* 2023, **14**:958.
  47. Phungula SM, Krüger K, Nofemela RS, Weldon CW: **Developmental diet, life stage and thermal acclimation affect thermal tolerance of the fall armyworm, *Spodoptera frugiperda*.** *Physiol Entomol* 2023, **48**:122-131.
- This study measured CTmax and CTmin of different life stages of fall army worms (*Spodoptera frugiperda*) reared on different diets and at diverse acclimation temperatures. Life stage had a large effect on both measures of thermal tolerance. Adults had lower CTmax compared to sixth and first instars, whereas the opposite was true for CTmin. Higher thermal tolerance of first instars is hypothesised to be attributed to microclimates (surface of leaves) which experience higher temperatures.
48. Gonzalez VH, Manweiler R, Smith AR, Oyen K, Cardona D, Wcislo WT: **Low heat tolerance and high desiccation resistance in nocturnal bees and the implications for nocturnal pollination under climate change.** *Sci Rep* 2023, **13**:22320.
  49. Bestelmeyer BT: **The trade-off between thermal tolerance and behavioural dominance in a subtropical South American ant community.** *J Anim Ecol* 2000, **69**:998-1009.
  50. Cerdà X, Retana J, Cros S: **Critical thermal limits in Mediterranean ant species: trade-off between mortality risk and foraging performance.** *Funct Ecol* 1998, **12**:45-55.
  51. Braschler B, Duffy GA, Nortje E, Kritzing-Klopper S, du Plessis D, Karenyi N, Leihy RI, Chown SL: **Realised rather than fundamental thermal niches predict site occupancy: implications for climate change forecasting.** *J Anim Ecol* 2020, **89**:2863-2875.
  52. Orlinick BL, Smith A, Medley KA, Westby KM: **Genetically based variation in heat tolerance covaries with climate in a globally important disease vector.** *Front Ecol Evol* 2024, **11**:1248673.
- This study measures CTmax for eight different populations of *Aedes albopictus* (mosquitoes) across four climate zones in the United States. They observed dramatic differences in CTmax across life stages and populations, with CTmax being higher for populations with higher precipitation and for larvae (a difference of 5.6–8.4°C from adults).
53. Li Y-J, Chen S-Y, Jørgensen LB, Overgaard J, Renault D, Colinet H, Ma C-S: **Interspecific differences in thermal tolerance landscape explain aphid community abundance under climate change.** *J Therm Biol* 2023, **114**:103583.
- This study measured the thermal tolerance and sensitivity of three species of co-existing cereal crop pests, *Sitobion avenae*, *Rhopalosiphum padi* and *Metopolophium dirhodum* using the thermal death time framework at three life stages (2, 6 and 12 days). Life stage differences in thermal tolerance and sensitivity were found to vary across species, but CTmax (1hr) only varied between 0.49 and 0.79°C among life stages across the species.
54. Gray EM: **Thermal acclimation in a complex life cycle: the effects of larval and adult thermal conditions on metabolic rate and heat resistance in *Culex pipiens* (Diptera: Culicidae).** *J Insect Physiol* 2013, **59**:1001-1007.
  55. Bennett JM, Calosi P, Clusella-Trullas S, Martínez B, Sunday J, Algar AC, Araújo MB, Hawkins BA, Keith S, Kühn I, et al.: **GlobTherm, a global database on thermal tolerances for aquatic and terrestrial organisms.** *Sci Data* 2018, **5**:180022.
  56. Pottier P, Wu NC, Earhart ML, Lagisz M, Alter K, Angelakopoulos R, Chakraborty A, Cowan Z-L, Killen SS, McCoy JCS, et al.: **Embryos are largely understudied in a representative sample of journals in conservation physiology.** *Conserv Physiol* 2026, **14**:coag006.
- This systematic review demonstrates a pronounced bias in research towards adult life stages, with embryos, larvae and juveniles being underrepresented (embryos 8–9% and juveniles 29% of studies), with insects making up 19% of studies reviewed.
57. Arnold PA, Noble DWA, Nicotra AB, Kearney MR, Rezende EL, Andrew SC, Briceño VF, Buckley LB, Christian KA, Clusella-Trullas S, et al.: **A framework for modelling thermal load sensitivity across life.** *Glob Change Biol* 2025, **31**:e70315.
- This review paper describes how to model thermal load sensitivity across animals and plants. It emphasises the need to incorporate an understanding of thermal exposure and the differing sensitivities of organisms and tissues to predict lethal and sub-lethal impacts of temperature. It describes how thermal load is a balance between damage

and repair to physiological processes and presents an updated thermal load sensitivity model with repair incorporated.

58. Buckley LB, Huey RB, Ma C-S: **How damage, recovery, and repair alter the fitness impacts of thermal stress.** *Integr Comp Biol* 2025, **65**:1061-1075.

This study uses aphids to show that rates of damage and repair in response to heat stress impact fecundity and development through time. Experiments demonstrate that temperatures above  $T_{opt}$ , with limited opportunities for recovery, decrease fitness. Models incorporating recovery better match experimental results and findings largely support other studies showing that increased nighttime temperatures above  $T_{opt}$  compromise fitness by limiting opportunities for recovery.

59. Jørgensen LB, Ørsted M, Malte H, Wang T, Overgaard J: **Extreme escalation of heat failure rates in ectotherms with global warming.** *Nature* 2022, **611**:93-98.

60. Vasudeva R: **Experimental evidence for stronger impacts of larval but not adult rearing temperature on female fertility and lifespan in a seed beetle.** *Evol Ecol* 2023, **37**:545-567.

This study explored how rearing temperatures (17, 25, 27 and 33 °C) experienced during larval and adult stages impact female reproduction and lifespan in the beetle, *Callosobruchus maculatus*. Stage-specific plasticity in lifespan and reproduction were observed with larval rearing temperatures having the greatest impact on reproduction. Fecundity and fertility were reduced most when larvae experienced 17°C whereas lifespan was maximised at 17°C, irrespective of the life stage.

61. Meena A, Maggu K, De Nardo AN, Sbilordo SH, Eggs B, Al Toma Sho R, Lüpold S: **Life stage-specific effects of heat stress on spermatogenesis and oogenesis in *Drosophila melanogaster*.** *J Therm Biol* 2024, **125**:104001.

This study examined how sublethal heat stress experienced at different life stages affects male reproductive performance. The authors found that heat stress during both development and adulthood had long-lasting negative effects on male fertility. These effects were linked to disrupted sperm production, smaller seminal vesicles and reduced sperm viability in flies reared at warmer temperatures.

62. Porcelli D, Gaston KJ, Butlin RK, Snook RR: **Local adaptation of reproductive performance during thermal stress.** *J Evol Biol* 2017, **30**:422-429.

63. Vives-Inglá M, Sala-García J, Stefanescu C, Casadó-Tortosa A, García M, Peñuelas J, Carnicer J: **Interspecific differences in microhabitat use expose insects to contrasting thermal mortality.** *Ecol Monogr* 2023, **93**:e1561.

64. Buckley LB, Arakaki AJ, Cannistra AF, Kharouba HM, Kingsolver JG: **Insect development, thermal plasticity and fitness**

**implications in changing, seasonal environments.** *Integr Comp Biol* 2017, **57**:988-998.

65. von Schmalensee L, Süess P, Roberts KT, Gotthard K, Lehmann P: **A quantitative model of temperature-dependent diapause progression.** *Proc Nat Acad Sci* 2024, **121**:e2407057121.

This study develops a quantitative model to explain diapause and uses the butterfly, *Pieris napi*, to validate the model. They show that diapause termination and post-diapause development in *P. napi* are directly sequential with different thermal maxima — diapause has maximal rates at low temperatures, whereas post-diapause maxima are at higher temperatures — allowing predictions of when diapause ends and how fast development proceeds post-diapause.

66. Nufio CR, Sheffer MM, Smith JM, Troutman MT, Bawa SJ, Taylor ED, Schoville SD, Williams CM, Buckley LB: **Insect size responses to climate change vary across elevations according to seasonal timing.** *PLoS Biol* 2025, **23**:e3002805.

This study leverages long-term data for six montane grasshoppers species along an altitudinal gradient to show: 1) size shifts have occurred based on elevation and species seasonal timing of development since the 1960s; 2) Size shifts have been greater for species at low elevations with early emerging species (overwintering as juveniles) increasing in size whereas the opposite is true for late emerging species and 3) interannual variation in temperature accounts for size shifts. Results suggest early-season species can take advantage of warming conditions to accelerate growth, whereas late-emerging species experience sub-optimal temperatures, which slow growth.

67. Fenberg PB, Self A, Stewart JR, Wilson RJ, Brooks SJ: **Exploring the universal ecological responses to climate change in a univoltine butterfly.** *J Anim Ecol* 2016, **85**:739-748.

68. Macgregor CJ, Thomas CD, Roy DB, Beaumont MA, Bell JR, Brereton T, Bridle JR, Dytham C, Fox R, Gotthard K, et al.: **Climate-induced phenology shifts linked to range expansions in species with multiple reproductive cycles per year.** *Nat Commun* 2019, **10**:4455.

69. Noer NK, Nielsen KL, Sverrisdóttir E, Kristensen TN, Bahrndorff S: **Temporal regulation of temperature tolerances and gene expression in an arctic insect.** *J Exp Biol* 2023, **226**:jeb245097.

This study highlights the importance of rapid plastic responses in thermal tolerance in an Arctic insect (the seed bug). Notably, recent thermal exposure, within just a few hours, affected both heat and cold tolerance in the field. Transcriptional responses were also highly sensitive to daily temperature fluctuations, with warmer conditions driving the largest shifts in gene expression and overall transcriptomic profiles.